

Ayush Sharma

Indian Institute of Technology, Goa

4th Year Undergraduate, **Mathematics and Computing**

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Education

B.Tech, Mathematics and Computing, IIT Goa,

2023 – Present

Experience

Software Development Intern at Cogniway

May 2025 – June 2026(Remote)

- Designed and implemented an automated **Therapy Session Scheduler** satisfying therapist availability, leave schedules, qualification matching, workload balancing, conflict-free allocation, and session spacing constraints.
- Built REST APIs, MongoDB models, and an AI-assisted Goal Management module which uses Gemini AI for hierarchical therapy planning, customizable assessment scales, therapist approval workflows, and patient progress tracking.

Projects

Cache Simulator [\[GitHub\]](#)

Sept 2025 – Nov 2025

- Situation/Task:** Built a configurable two-level set-associative cache simulator in C++ to model and quantify how cache design decisions affect memory performance across different workload patterns.
- Action:** Implemented O(1) LRU replacement, write-back/write-through, write-allocate/no-write-allocate, and next-line prefetching using STL data structures.
- Accomplishments:** Reduced cache misses by up to 50% on sequential traces using next-line prefetching, validated correctness across 8 manually derived test cases (AMAT, hit rate, writebacks).
- Tech Stack/Concepts:** C++, Cache Design, STL

Task Queue System [\[GitHub\]](#)

Dec 2025 – June 2026

- Situation/Task:** Required a production-grade asynchronous task execution system with reliable delivery, fault tolerance, and duplicate prevention.
- Action:** Built a distributed task queue using Node.js, Express.js, PostgreSQL, and Redis with multi-layer idempotency, exponential backoff retries, zombie task recovery, and Redis-backed queue management for reliable asynchronous processing.
- Accomplishments:** Achieved ~2K tasks/sec enqueue throughput with 20 ms average API latency (53 ms p99), verified exactly-once execution across 1.5K+ tasks using 4 concurrent workers, and eliminated a TOCTOU race by replacing a check-then-insert workflow with an atomic database operation.
- Tech Stack:** Node.js, Express.js, PostgreSQL, Redis

Secure Endpoint Monitoring Platform

May 2026 – Present

- Situation/Task:** Design a distributed Linux endpoint monitoring platform for labs with centralized orchestration and tamper-aware activity collection.
- Action:** Building a trigger-driven client-server architecture using Bash, SSH, iptables, and Linux utilities to coordinate process, filesystem, terminal, Vim, and network monitoring across multiple clients.
- Accomplishments:** Session-based logging, automated log collection and analysis, network isolation, suspicious activity detection for accurate and fair analysis.
- Tech Stack/Concepts:** Linux, Bash, Python, iptables

Skills

Languages: C++, Python, JavaScript

Backend: Node.js, Express.js, PostgreSQL, Redis, MongoDB, REST APIs, JWT Authentication

Tools: Git, Postman, Vercel, Render

Technical Concepts: DSA, Operating Systems, Database Management Systems, Computer Architecture, Computer Networks, Memory Hierarchy, Linux, OOPS

Achievements & Extracurricular

- Achieved LeetCode rating of 1900+ (top 5% globally), demonstrating problem-solving and DSA proficiency.
- Secured top 32 position among 100+ teams at Goa Police Hackathon 2024 with a Fraud Detection System achieving 85%+ accuracy.
- Represented IIT Goa at Inter-IIT Basketball Championship.